

LLM-as-music-judge

Can LLM act as a “music judge” that listens, reasons, and scores generative audio like a human would?

CNMAT Group 2

Editor: Song-Ze Yu

Question Statement

Abstract

- Nowadays, generative AI and large language models (LLMs) are frequently used in music generation tasks.
- However, most systems still lack a fair and explainable evaluation process for assessing the quality or **emotional** fidelity of generated music.

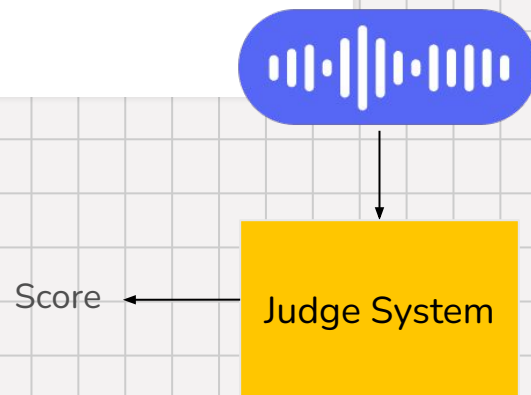
Related Works

- CLAP: Enable text-to-audio **multimodal** alignment, but
 - (1) focus only on representation similarity, not evaluative reasoning
 - (2) lacks fine-grained evaluation of musical elements such as tone, rhythm, or expressiveness.
- MERT: Music embedding to represent **emotional or textual alignment**, but cannot directly output an interpretable score or critique.

Research Abstract

Create an fair evaluation system for music, used for re-prompting LLM generation of a tone or music.

Unlike prior embedding models that only measure similarity, our framework allows the LLM to reason about tone design and musical quality, effectively transforming static embeddings into actionable evaluation.

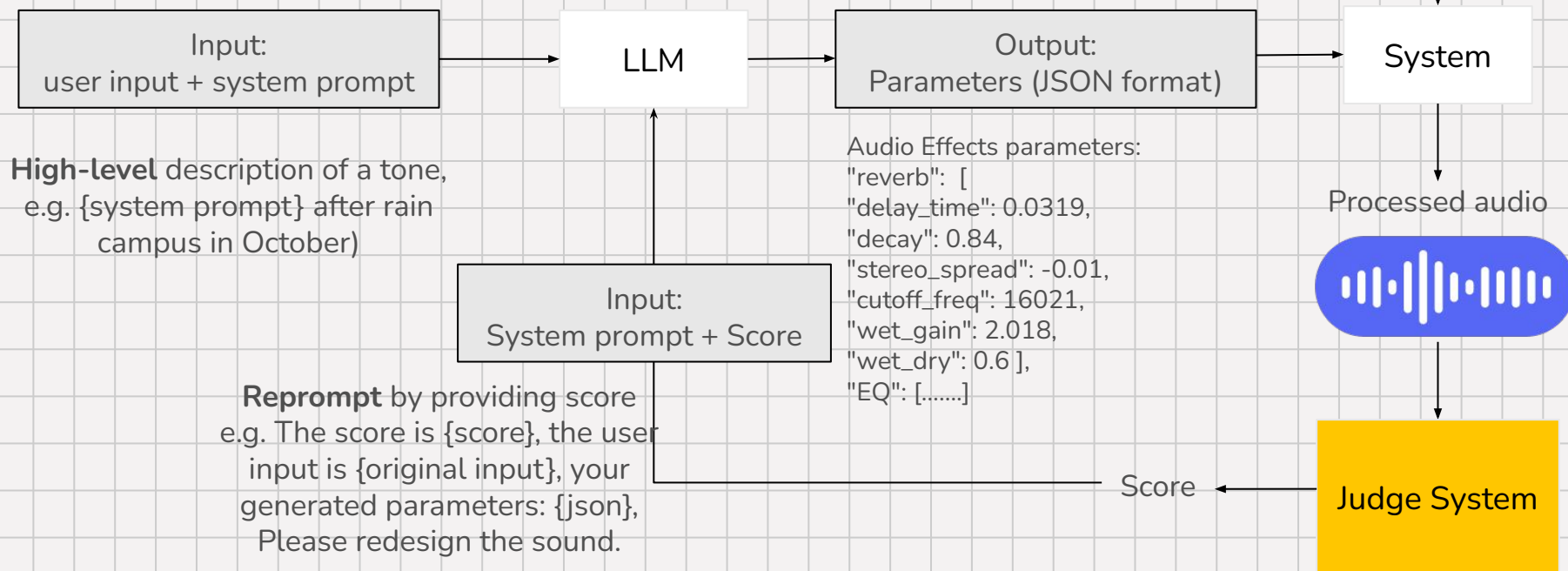


Experiment Method



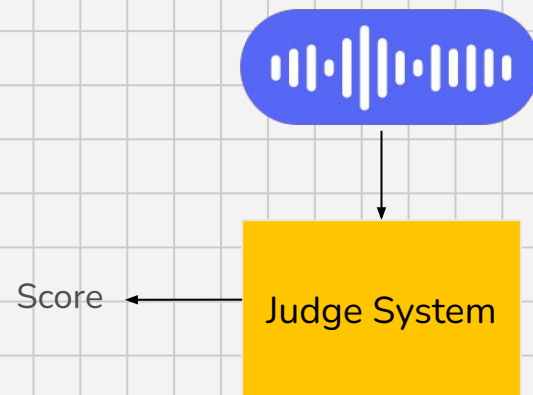
A **closed-loop** architecture enabling LLMs to iteratively **refine tone design** based on **self-evaluated scores**.

Experimental Architecture



Architecture Demo

This demo will be our testing baseline for our experiment.



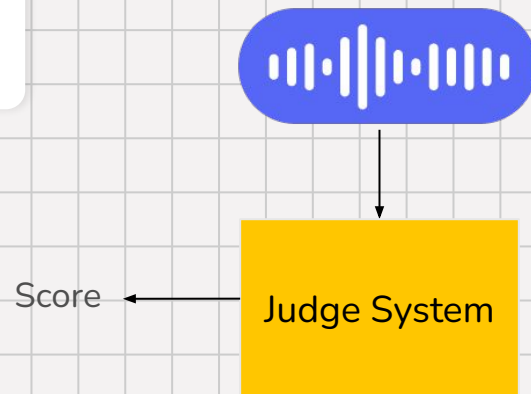
Judge System - *Our Main Contribution*



Traditional evaluation (e.g., CLAP, MERT, FAD) → measure similarity, but music quality and emotion are subjective and multidimensional.

We need a qualitative scoring framework that captures “vibes,” not just accuracy

Related work: [VibeCheck - ICLR 2025](#)



Recap...

Combining three different components

$$R = \alpha \cdot R_{\text{ToneJudge}} + \beta \cdot R_{\text{AudioScore}} + \gamma \cdot R_{\text{NIMA}}$$

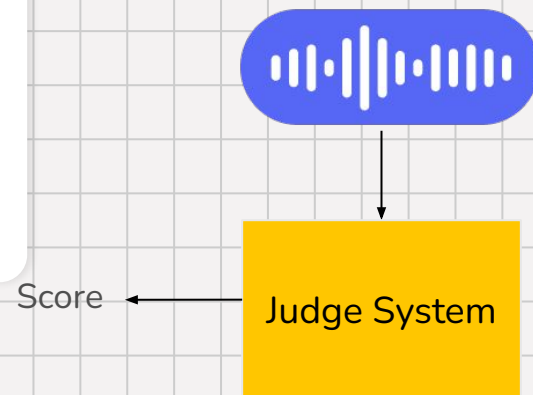
Reference paper: agent with RL

(<https://aimc2024.pubpub.org/pub/9zd8yyyyv/release/1>)

Judge System - *Our Main Contribution*

Each score = combination of **quantitative** and **qualitative** components:

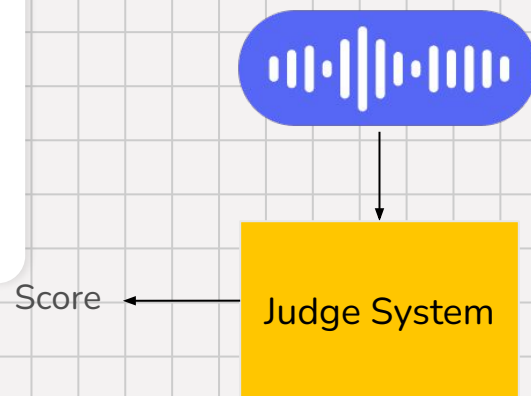
Type	Source	Example Dimension	Description
Quantitative	DSP metrics	Loudness, spectral balance	Objective physical features
Qualitative (Vibe)	LLM reasoning	Calm -> Energetic; Warm -> Bright; Natural -> Synthetic	From LLM-as-judge
Preference Alignment	User real data	Warm ->	Used for calibration of vibe weights



Judge System - Vibe part



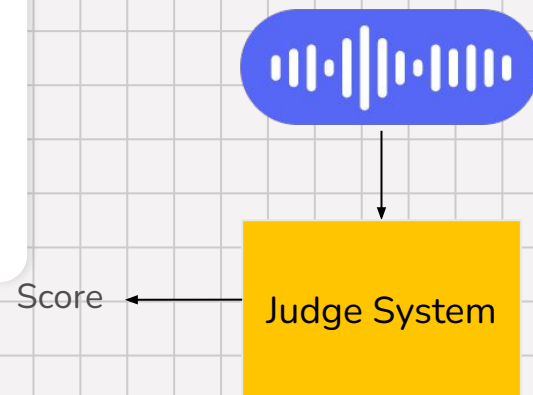
1. Collect comparisons: feed two generated audios to the LLM-judge.
2. Ask LLM to describe the difference in natural language (e.g., “Track A feels more melancholic and natural”).
3. Extract vibe axes automatically:
 - “Mood: sad → joyful”
 - “Texture: synthetic → organic”
4. Quantify each vibe into numeric sub-scores (-1, 0, 1).
5. Aggregate weighted score based on task goal or user preference.



Judge System - Vibe part

Metric of Success:

Metric	Meaning
Well-defined	Do LLMs agree on the same vibe for the same music pair?
Differentiating	Does the vibe axis clearly distinguish different music systems?
User-aligned	Do vibe-based scores correlate with human emotional ratings?



Research Milestone

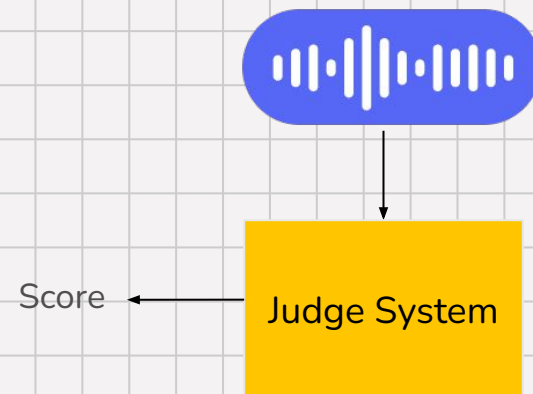
2025

- Baseline system : 2025/10
 - Minimal self-loop between LLM → Parameter → Audio → Judge (Jimmy)
 - Find dataset and metrics related
 - CAL500
 - Research and Design the “judge”

- Judge System (Core): 2025/11

.....

2026



Thank you

Reference: VibeCheck

ABSTRACT

Large language models (LLMs) often exhibit subtle yet distinctive characteristics in their outputs that users intuitively recognize, but struggle to quantify. These "vibes" – such as tone, formatting, or writing style – influence user preferences, yet traditional evaluations focus primarily on the singular vibe of correctness. We introduce **VibeCheck**, a system for automatically comparing a pair of LLMs by discovering identifying traits of a model ("vibes") that are well-defined, differentiating, and user-aligned. VibeCheck iteratively discovers vibes from model outputs and then utilizes a panel of LLM judges to quantitatively measure the utility of each vibe. We validate that the vibes generated by VibeCheck align with those found in human discovery and run VibeCheck on pairwise preference data from real-world user conversations with Llama-3-70b vs GPT-4. VibeCheck reveals that Llama has a friendly, funny, and somewhat controversial vibe. These vibes predict model identity with 80% accuracy and human preference with 61% accuracy. Lastly, we run VibeCheck on a variety of models and tasks including summarization, math, and captioning to provide insight into differences in model behavior. VibeCheck discovers vibes like Command X prefers to add concrete intros and conclusions when summarizing in comparison to TNGl, Llama-405b often overexplains its thought process on math problems compared to GPT-4o, and GPT-4 prefers to focus on the mood and emotions of the scene when captioning compared to Gemini-1.5-Flash. Code and vibe visualizer found at <https://bench-mark.org/>

Problem to solve:

Nowadays LLM evaluation method focus on measure accuracy, lack of open-ended nature of LLM applications.

e.g.
tone and creativity -> writing;
efficiency and readability -> coding

What is a vibe?

Prompt: What is the best coffee?

Output A: **What a bold question!** After considering various factors, I declare...

Output B: Identifying the "best" coffee is challenging because taste is subjective...

"On what axes do these two outputs differ?"

Vibe (low → high)

Friendliness
formal → friendly

How do we score vibes?

Prompt: If I was a mouse ..

Output A: If you were a mouse, we'd find a way to communicate effectively...

Output B: Ahahaha! Oh, what a delightful pun!

Judge *i*

"Which output ranks **higher** on the **friendliness** axis?
Respond with A, B, or equal"

$$V_i(p_1, O_a, O_b) = B = -1$$

How do we quantify vibe utility?

Judge 1	Judge 2	Avg Score	Preference
$V_1(p_1, O_A, O_B)$	$V_2(p_1, O_A, O_B)$	1	A
$V_1(p_2, O_A, O_B)$	$V_2(p_2, O_A, O_B)$	-1	B
⋮	⋮	⋮	⋮

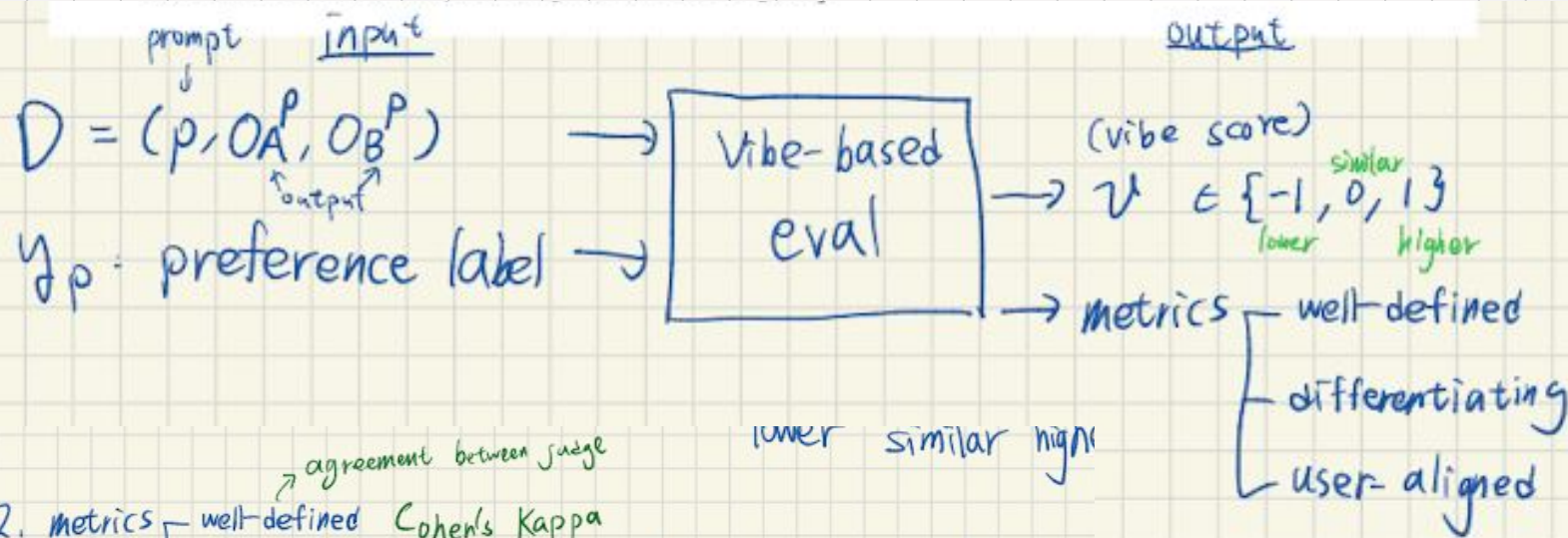
Well Defined → Agreement between Judge 1 & 2 → 0.4

Differentiating → Ability to predict model ID from friendliness → 55%

User-Aligned → Ability to predict preference from friendliness → 55%

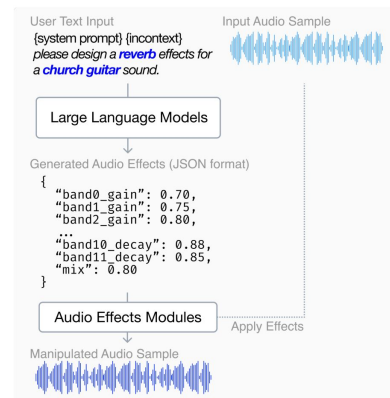
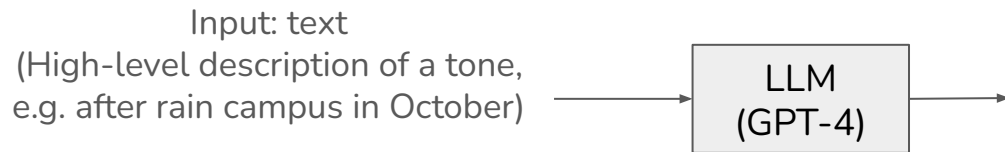
Solution

[Back to our proposal score design](#)



model matching accuracy

System Architecture



$R_{\text{ToneJudge}}$: semantic match

Reference paper: CLAIR-A, CLAP

Ref: CLAIR-A

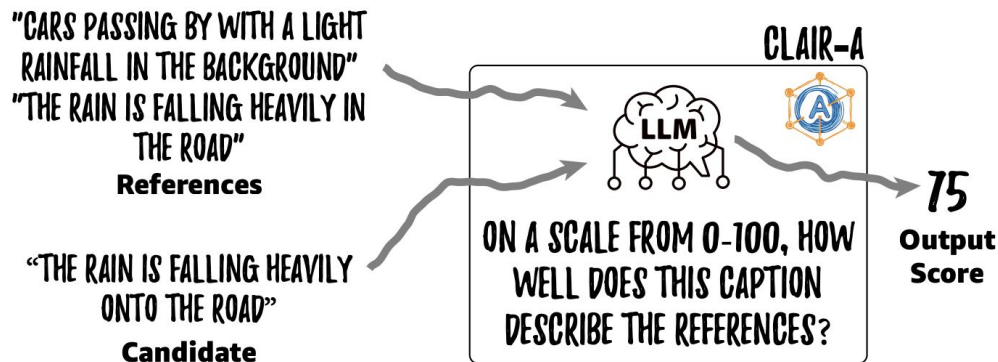


Fig. 1: CLAIR_A: a simple, domain-specific, measure for evaluating audio captions. By simply leveraging large language models, in-context learning, and guided generation techniques, CLAIR_A both correlates strongly with human judgments of caption quality and produces both *high-quality* and *interpretable* scores according to human raters.

You are tasked with evaluating if a set of candidate captions accurately describes the same sound in a video clip as a reference set of captions. Start by assessing the accuracy and precision of how the audio characteristics are captured in the captions, scoring from 0 to 90 based on this aspect alone. After this initial assessment, you may add additional points (from 0 to 10) based on the quality of grammar and the detailed, reasonable descriptions present in the captions.

Candidate set:

{candidate captions}

Reference set:

{reference captions}

Combine these two aspects for a final evaluation score on a scale from 0 to 100, reflecting the likelihood that the candidate set is describing the same sound as the reference set. Format your response in JSON with a key "score", value between 0 and 100, and a key "reason" with a string value explaining your assessment.

Fig. 2: The prompt used for CLAIR_A. Instead of asking for a single score, we find that a multi-tiered scoring system, which allocates points on a rubric, can mitigate ties and improve correlation on low-quality samples in the audio domain.

· $R_{\text{AudioScore}}$: **audio fidelity**

- FAD: music distance
- CLAP / MERT / VGGish embedding

Reference: FAD (<https://arxiv.org/abs/1812.08466>)

MAD (?)

$\gamma \cdot R_{\text{NIMA}}$: preference

Reference paper: NIMA

Target:

审美偏好

Reference paper: NIMA

Metrics

Objective

Emotion Recognition Accuracy (ERA)

Accuracy/F1-score of a pre-trained classifier (e.g., CLAP) on generated audio vs. ground-truth labels.

Measures semantic fidelity; benchmark against human music.

Objective

Audio-Text Alignment Score

Cosine similarity between CLAP embeddings of audio and emotion prompt.

Quantifies how well "sad" prompt matches output timbre/melody.

Objective

Statistical Divergence

KL-divergence on emotion-induced features (e.g., tempo for happiness).

Compares distributions to emotional baselines.

Subjective - need people to listen (unrealistic to get a lot of data)

Mean Opinion Score (MOS)

Likert-scale ratings (1-5) from listeners on emotional evocation.

Captures perceptual impact; use A/B tests vs. human music.

Hybrid

BLEU/METEOR for Symbolic

Sequence alignment if outputting MIDI; adapt for emotional motifs.

For symbolic generation; scores motif-emotion match.

CLAP

- Multimodal model for aligning audio and text embeddings via contrastive learning.
- Pretrained on large-scale audio-text pairs (e.g., AudioSet, Clotho).
- Key: Zero-shot retrieval – query audio with text like "joyful piano" to match clips.

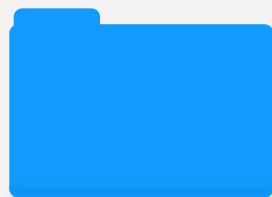
Computes cosine similarity between audio features and emotion descriptors

Applications: Music tagging, captioning; achieves ~70% accuracy in emotion classification on datasets like GTZAN.

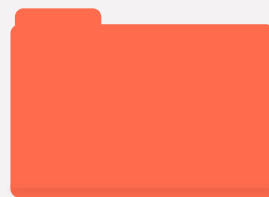
- **Implementation:**
 - **Step 1:** Extract audio embeddings from LLM-generated clips (e.g., via Hugging Face CLAP model: `clap_model.encode_audio(audio_waveform)`).
 - **Step 2:** Align with emotion prompts: Compute similarity = `cosine(CLAP_audio_emb, CLAP_text_emb("happy melody"))`.
 - **Step 3:** Threshold-based scoring (>0.7 = strong match) for metrics like Audio-Text Alignment Score.



About



Introduction

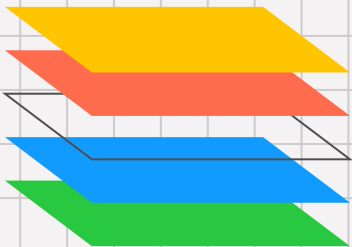
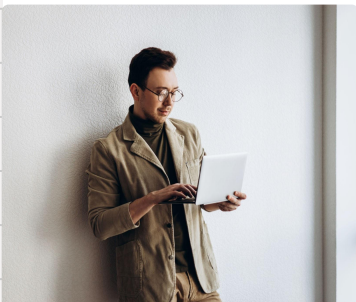


Class



Contact

About Me



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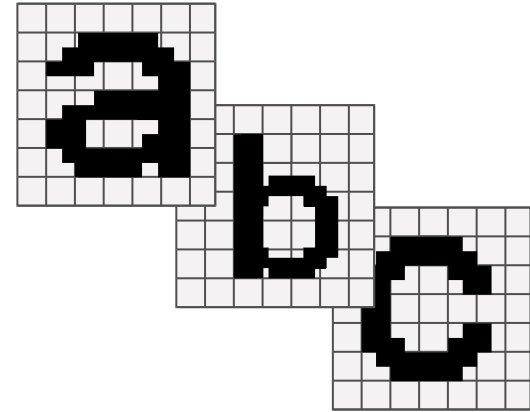
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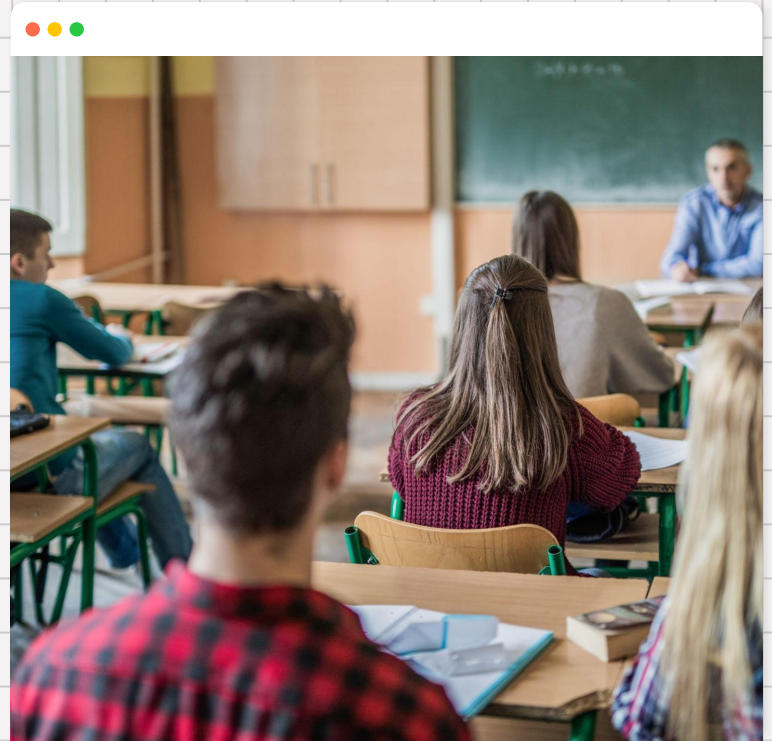
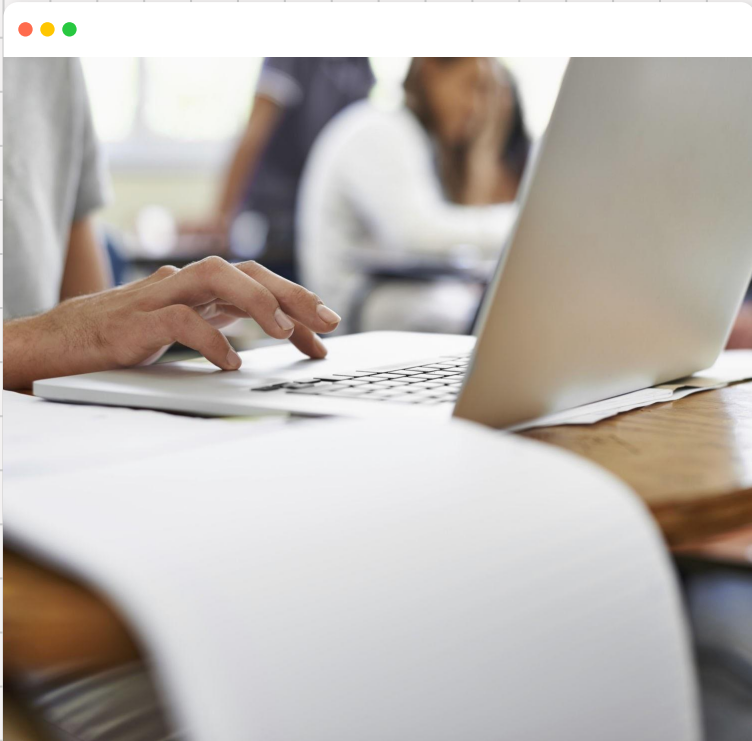
Learning Outcome

Outline

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Class Methods



Class Expectations

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Schedule

Date

Topic

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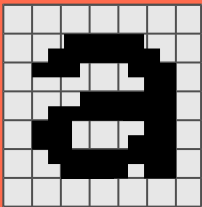
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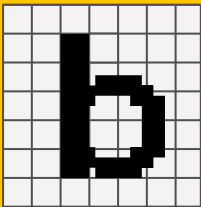
Tips

Class Materials



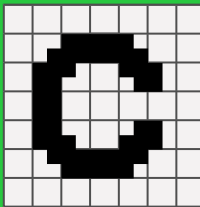
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Contact Me

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